



Utilizing Pseudo Color Image to Improve the Performance of Deep Transfer Learning–Based Computer-Aided Diagnosis Schemes in Breast Mass Classification

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Abstract

The purpose of this study is to investigate the impact of using morphological information in classifying suspicious breast lesions. The widespread use of deep transfer learning can significantly improve the performance of the mammogram based CADx schemes. However, digital mammograms are grayscale images, while deep learning models are typically optimized using the natural images containing three channels. Thus, it is needed to convert the grayscale mammograms into three channel images for the input of deep transfer models. This study aims to develop a novel pseudo color image generation method which utilizes the mass contour information to enhance the classification performance. Accordingly, a total of 830 breast cancer cases were retrospectively collected, which contains 310 benign and 520 malignant cases, respectively. For each case, a total of four regions of interest (ROI) are collected from the grayscale images captured for both the CC and MLO views of the two breasts. Meanwhile, a total of seven pseudo color image sets are generated as the input of the deep learning models, which are created through a combination of the original grayscale image, a histogram equalized image, a bilaterally filtered image, and a segmented mass. Accordingly, the output features from four identical pre-trained deep learning models are concatenated and then processed by a support vector machine–based classifier to generate the final benign/malignant labels. The performance of each image set was evaluated and compared. The results demonstrate that the pseudo color sets containing the manually segmented mass performed significantly better than all other pseudo color sets, which achieved an AUC (area under the ROC curve) up to 0.889 ± 0.012 and an overall accuracy up to 0.816 ± 0.020 , respectively. At the same time, the performance improvement is also dependent on the accuracy of the mass segmentation. The results of this study support our hypothesis that adding accurately segmented mass contours can provide complementary information, thereby enhancing the performance of the deep transfer model in classifying suspicious breast lesions.

Keywords Computer-aided diagnosis (CAD) · Pseudo color image generation · Deep transfer learning · Breast cancer · Breast lesion classification · Multi-view image feature analysis · Radiomics features · Image feature fusion

Introduction

Breast cancer is one of the leading causes of death in women worldwide [1], and mammography is one of the most effective tools to screen the large women population

and identify breast cancer patients at an early stage. Despite the significance and importance, interpreting mammography images is a challenging task, as radiologists will analyze a total of four images (i.e., craniocaudal [CC] view and mediolateral oblique (MLO) view for each breast) to determine if there are any suspicious regions that must be biopsied. Therefore, the visual examination and interpretation of digital mammograms have a high false-positive rate [2].

To address this clinical challenge, many computer-aided diagnosis (CADx) systems have been developed to assist radiologists to classify suspicious lesions [3, 4]. Especially, deep learning [5, 6]–based CADx schemes use convolutional neural networks (CNNs) or transformers to automatically

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classify a suspicious lesion from the input image. Since these networks learn image features directly from the suspicious lesion, they can identify patterns relevant to the target domain that cannot be recognized by the radiologists. In many studies, the deep learning–based CADx schemes demonstrate superior performance [5, 6]. However, in order to achieve high performance, a large and diverse dataset is required to sufficiently optimize the deep networks, which is often unavailable in the medical imaging domain. Transfer learning has thus emerged as a popular solution [7], in which a network is first pre-trained on a large, non-medical dataset, and then fine-tuned on the medical task using a small medical dataset.

For transfer learning, the pre-trained deep learning models are usually optimized on a large dataset of color images with three channels (e.g., ImageNet) [5, 8], while the following mammogram dataset contains single-channel grayscale images. Although mammogram images can be simply stacked as input to the deep network [9–11], pseudo color augmentation techniques provide additional and complementary information to the deep learning model, and therefore the well-trained model can yield improved classification and detection performance. For example, Min et al. [12] created pseudo color ROIs for mass detection task, which placed the original image in the red channel and generated two additional images using a multi-scale morphological sifter (MMS) for the green and blue channels. The MMS was developed in a manner that aimed to extract spicules, as malignant lesions tend to have spiculated margins. The pseudo color ROI is then used as input in a pre-trained Mask R-CNN network with a ResNet101 backbone for mass detection and segmentation. Pseudo color image based model achieved an average sensitivity of 0.9 (at 0.90 FPI, false positive per image), which outperformed the single-channel ROI based model. In addition, Jones et al. [13] evaluated the performance of a pseudo color image sets created by stacking the original grayscale image, a bilaterally filtered image, and a histogram equalized version. When using the sample deep learning model (VGG-16), the proposed pseudo image set achieved a better performance than stacking three original grayscale images in classifying craniocaudal (CC) view breast lesions.

Despite the success of these previous studies [12–16], very few research has been performed on adding complementary mass contour information. It is known that malignant and benign masses can often be distinguished based on their contours as benign masses tend to have round or oval contours, while malignant masses tend to be irregular in shape with highly spiculated margins [17, 18]. We hypothesize that adding a fully segmented mass that captures these morphological information may aid in classifying suspicious breast lesions by providing increased information. In this study, we compare the performance of seven pseudo color

image sets as inputs to a multi-view CADx scheme for classifying suspicious breast lesions as malignant or benign. All the details are presented as follows.

Materials and Methods

Database

The dataset consists of full-field digital mammograms (FFDM) that were acquired under an institutional review board (IRB) approved protocol. All the mammograms were acquired using Hologic Selenia digital mammography machines (Hologic Inc., Bedford, MA, USA). Details pertaining to the image specifics can be found in several of our previous publications [13, 19, 20]. In this study, we retrospectively assembled a dataset by selecting cases that contained all four images taken during the screening mammography exam (left and right CC and MLO images). We only select cases where a mass can be seen in both the CC and MLO view of the same breast. An experienced radiologist has marked the center of each lesion, and all suspicious lesions have been biopsy proven as malignant or benign. The center of each lesion was used as a reference to manually annotate the suspicious mass boundaries. The annotations were converted into binary segmentation masks and treated as the ground truth.

Given that the original mammogram has a very high resolution (4090×3080) and the mass lesion detection is usually conducted on the subsampled images, we first applied a 5×5 average kernel on the mammogram to downsample the image to a size of 818×616 . As some cases contain multiple masses in the same breast, we conducted an ipsilateral matching scheme to ensure that masses in each view were properly matched up prior to classification. After ipsilateral matching, each mass is represented by four images, the CC and MLO view of the breast that contains the suspicious mass and the CC and MLO view of the contralateral breast. In this study, we will quantify the bilateral asymmetry between breasts as conducted in our previous work [21]. To accomplish this, we first need to register the contralateral breast images to the image that contains the suspicious mass. The bilateral registration scheme is conducted using a multi-resolution B-spline transformation that optimizes the Mattes mutual information metric. The FFDM image containing the suspicious mass is selected as the fixed image, and the contralateral image is selected as the moving image. Thus the registration process will not distort the location of the mass that has been marked by a radiologist. In-depth details pertaining to the ipsilateral matching and bilateral registration schemes can be found in our previous study [21]. After bilateral registration, a 64×64 pixel region of interest (ROIs) is extracted surrounding the center of the lesion. Since only

the CC and MLO images of the breast containing the lesion have been center marked, we extract the same ROI from the contralateral breast. As the images have been bilaterally registered, they should represent the same area in the breast. A visual representation of this process is depicted in Fig. 1 step 1. In this figure, the suspicious lesions are highlighted in the red bounding boxes in the left breast.

There were 1065 cases containing four FFDM images representing the left and right CC and MLO view. After removing the cases which misses the annotating information or fail in preprocessing step, the final dataset contains a total of 830 cases, among which 310 cases are biopsy-proven benign and 520 cases are biopsy-proven malignant. This corresponds to 3320 64×64 grayscale ROIs and 1660 binary segmentation masks.

Automated Mass Segmentation

Prior to the creation of the pseudo color ROIs, a U-Net was trained to perform automatic mass segmentation. The U-Net was chosen for segmentation because it has demonstrated high performance in breast mass classification tasks [22–24] and is one of the most widely used models in tumor/mass segmentation [25]. The U-Net is composed of an encoding pathway and a decoding pathway. The encoding pathway learns abstract representations of the input image through convolutional operations and down-sampling via max pooling layers. The decoder pathway uses the abstract information learned from the encoder blocks to reconstruct the segmentation mask of the input via up-sampling and convolutional blocks. The network achieves this through skip connections which act as bridges between the corresponding encoder and decoder blocks, allowing spatial information to be preserved [26]. The encoding pathway and decoding pathway consist of two symmetrical halves of the network which give the network its “U” structure.

In this study, we adopted the standard U-Net architecture for the segmentation task, which consists of four down-sampling blocks and four up-sampling blocks [26]. For each block, a dropout layer with a probability of 0.2 is added after the first convolutional layer, aiming to improve the model robustness and generalizability. The output layer of the network uses a 1×1 convolution with a sigmoid activation function to generate the final probability map of each pixel belonging to either the foreground or background. To train the network, the Adam optimizer was used with an initial learning rate of 0.001 on mini batches of size 16 for 100 epochs. The learning rate decayed exponentially at a rate of e^{-1} after the first ten epochs. A custom loss function that combines binary cross entropy (BCE) loss and dice loss was used to emphasize capturing fine details such as spiculations:

$$L_{combined}(y_{true}, y_{pred}) = \alpha L_{Dice}(y_{true}, y_{pred}) + (1 - \alpha) L_{BCE}(y_{true}, y_{pred}) \quad (1)$$

$$L_{Dice}(y_{true}, y_{pred}) = 1 - \frac{2 \times |y_{true} \cap y_{pred}|}{|y_{true}| + |y_{pred}|} \quad (2)$$

$$L_{BCE}(y_{true}, y_{pred}) = -(y_{true} \times \log(y_{pred}) + (1 - y_{true}) \times \log(1 - y_{pred})) \quad (3)$$

where y_{true} is the binary ground truth image, y_{pred} is the predicted binary segmentation mask, and α is the weight assigned to each component in the loss term. Different values for α were investigated. The final value is set at 0.7. The manual segmentations are treated as the ground truth. During the training process, the fivefold cross-validation strategy was utilized. In each training cycle, 80% of the data is used for training (60% for training, 20% for validation), while 20% is reserved for testing. This is performed in a manner that each image appears once in the testing set; therefore, each image will have a corresponding segmentation mask produced by the U-Net which can be subsequently used to create the pseudo color image. To generate the segmented mass image, the binary mask produced by the U-Net will be applied to the original image, resulting in an image that contains only the segmented mass.

While there are four images associated with each mass, only two of them contain the suspicious mass. The binary segmentation mask will be applied to the contralateral image of the same projection view. For example, if the mass is visible in the LCC and LMLO images, the segmentation mask for the LCC image will be applied to the RCC image, and the LMLO segmentation mask will also be applied to the RMLO image. Since the contralateral breast images have been bilaterally registered to the images containing the suspicious mass, applying the binary segmentation masks in this manner will segment the corresponding area in the contralateral breast. The performance of the mass segmentation was evaluated using the dice similarity coefficient (DSC) and the Jaccard Index.

Pseudo Color Image Generation

We create seven variations of pseudo color ROIs by stacking different versions of the original ROI in three channels, thereby generating three channel images that mimic the three channel color images in the ImageNet dataset. Table 1 demonstrates all these different pseudo color image sets used in this study. In six of all the seven sets, we use a combination of the original single-channel image (I_o), a histogram equalized version (I_{HE}), a bilaterally filtered version (I_{BF}), and the segmented mass image (I_{seg}). The final set is created by mapping the single-channel grayscale ROI values to all

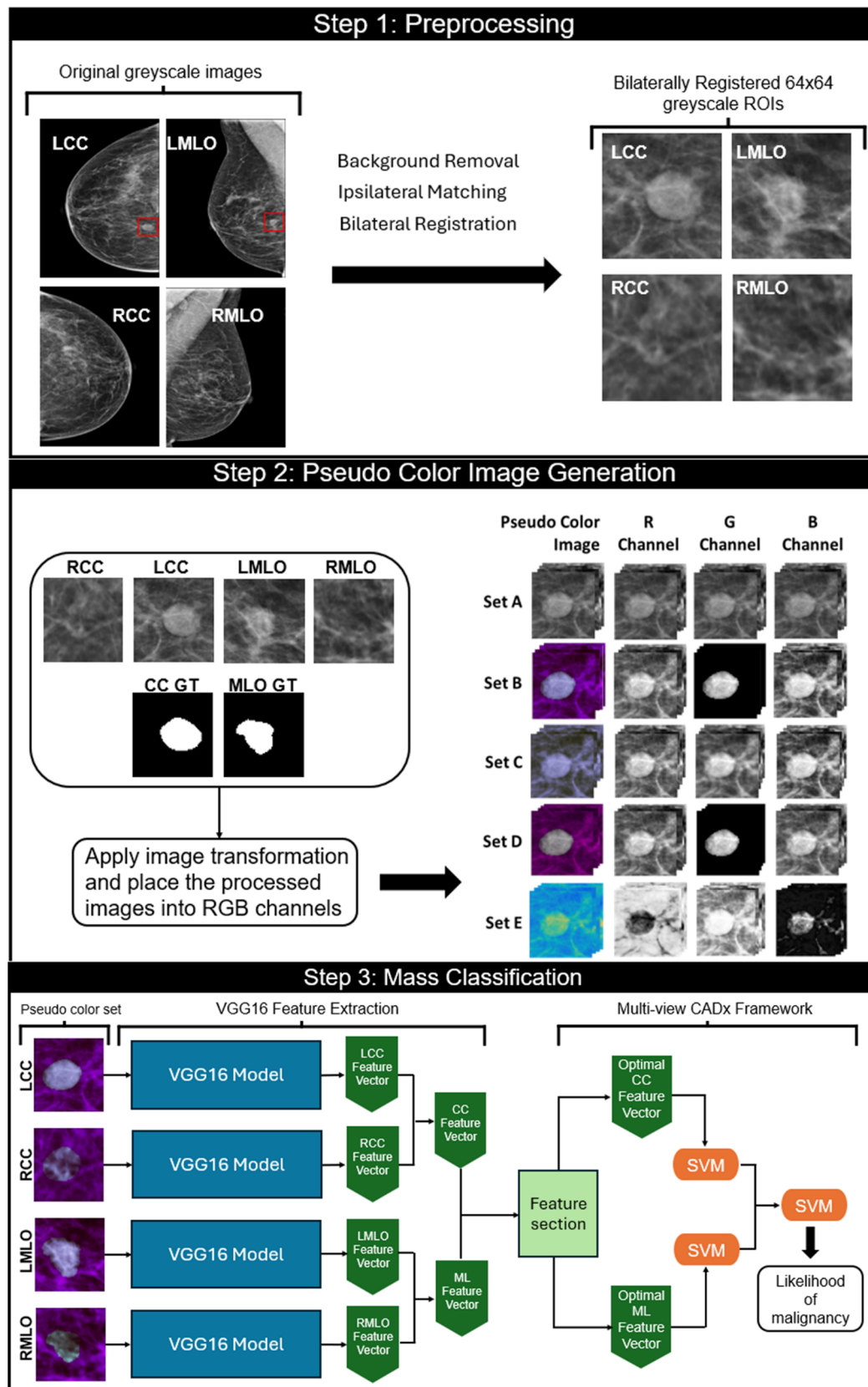


Fig. 1 A schematic diagram of this study

Table 1 Summary of pseudo color image sets

Set	R	G	B
A	I_o	I_o	I_o
B_{gt}	I_o	I_{seggt}	I_{he}
B_{unet}	I_o	$I_{segunet}$	I_{he}
C	I_o	I_{bf}	I_o
D_{gt}	I_o	I_{seggt}	I_o
D_{unet}	I_o	$I_{segunet}$	I_o
E	<i>Parula color mapped</i>		

Set E is created by applying the parula colormap to the original image

I_o original image, I_{he} histogram equalized variant, I_{bf} bilaterally filtered variant, I_{seggt} segmented mass using the ground truth, $I_{segunet}$ segmented mass using the U-Net predicted mask

available values in the parula colormap [27]. The histogram equalization preprocessing technique is selected to increase the contrast as mammography is an X-ray-based technique that inherently has low contrast. The bilateral filtering preprocessing technique is selected to de-noise the images as this technique is able to preserve edge and textural information while reducing noise [28].

To best investigate the effect of including a segmented mass image in the pseudo color images, we use both the ground truth segmentation mask and a U-Net-generated automated segmentation mask to create the pseudo color images. The motivation for using an automated segmentation step is to avoid the need for a manual segmentation step in future studies, as this is extremely time consuming and very user dependent [29]. Therefore, Set B and Set D are divided into two different sets, Set B ground truth (B_{gt}), which contains the ground truth segmentation, and Set B U-Net (B_{unet}), which contains the U-Net predicted segmentation mask. The same rule follows for Set D. As a result, a total of seven pseudo color sets were generated in this study.

Mass Classification

Recent research has demonstrated that multi-view CADx schemes tend to outperform single-view CADx schemes [30, 31]. Our previous study investigated the optimal method for fusing feature vectors from multiple views taken during a mammography exam [21]. In this study, the same pipeline was adopted, which involves extracting deep transfer learning-based features from each of the four mammography views, quantifying the bilateral asymmetry, and training a two-stage classification system which predicts the likelihood of a mass being malignant.

Table 2 VGG16 architecture used for feature extraction

Block	Layer	Size	Filter size
1	Convolution-1	$224 \times 224 \times 64$	3×3
	Convolution-2	$224 \times 224 \times 64$	3×3
	Max pooling	$112 \times 112 \times 64$	-
2	Convolution-1	$112 \times 112 \times 128$	3×3
	Convolution-2	$112 \times 112 \times 128$	3×3
	Max pooling	$56 \times 56 \times 128$	-
3	Convolution-1	$56 \times 56 \times 256$	3×3
	Convolution-2	$56 \times 56 \times 256$	3×3
	Convolution-3	$56 \times 56 \times 256$	3×3
	Max pooling	$28 \times 28 \times 256$	-
4	Convolution-1	$28 \times 28 \times 512$	3×3
	Convolution-2	$28 \times 28 \times 512$	3×3
	Convolution-3	$28 \times 28 \times 512$	3×3
	Max pooling	$14 \times 14 \times 512$	-
5	Convolution-1	$14 \times 14 \times 512$	3×3
	Convolution-2	$14 \times 14 \times 512$	3×3
	Convolution-3	$14 \times 14 \times 512$	3×3
	Max pooling	$7 \times 7 \times 512$	-

Transfer Learning and Feature Extraction

In this work, the VGG16 network was utilized. Through transfer learning, we can use the weights learned from the natural image classification task to classify suspicious breast masses as malignant or benign. Before using the pre-trained network, our input images are modified to match the ImageNet input image dimension of $224 \times 224 \times 3$. The original 64×64 pixel images are resized to 224×224 via bilinear interpolation and converted into three channel pseudo color images as described in the “Automated Mass Segmentation” section. The VGG16 network has a relatively simple architecture that consists of five convolutional blocks followed by several dense layers. Blocks 1 and 2 contain two convolutional layers followed by a max pooling layer, while blocks 3, 4, and 5 contain three convolutional layers followed by a max pooling layer [32]. In this study, we use the pre-trained VGG16 network as a feature extractor by freezing all weights and removing the top dense layers. The architecture adopted for this study can be seen in Table 2. Features are then extracted from all four images after the final max pooling layer and flattened into a 25,088-dimensional vector.

Multi-view CADx Framework

A multi-view CADx framework is used to classify each case as malignant or benign. After pseudo color image generation, there are seven image sets. Each set contains four subsets representing the left and right CC and MLO images of the mass. For each image in the set, deep learning features are extracted as one 25,088 feature vector. Each feature vector is then normalized from 0 to 1, and the bilateral asymmetry is quantified

by computing the absolute difference between the feature vectors representing the corresponding bilateral views (LCC-RCC, LMLO-RMLO). This results in two feature vectors, a CC and an MLO feature vector, each with a size of 25,088. To reduce the high dimensionality of these vectors, we first apply variance thresholding to each set using a threshold of 0.2. We then apply a sequential forward feature selection (SFFS) algorithm to obtain the optimal feature set [33]. The SFFS algorithm is conducted using a linear support vector machine (SVM) over fourfold cross-validation. The optimal CC and optimal MLO feature vectors are then used to train two SVMs independently. The output of each SVM is fused and used to train a final SVM. All SVMs are trained and tested using fivefold cross-validation. Since our dataset is slightly imbalanced, we embed the synthetic minority oversampling technique (SMOTE) into each fold to resample the minority cases to balance the training dataset.

To assess the performance of the mass classification, the likelihood score of a case being malignant is generated for each case to create a receiver operating characteristic (ROC) curve. The area under the ROC curve (AUC) are then computed, and Delong's test is used to evaluate statistical significance between the ROC curves of different pseudo color sets [34]. Meanwhile, five other metrics are also generated for performance evaluation, namely accuracy, sensitivity, specificity, precision, and F1 score.

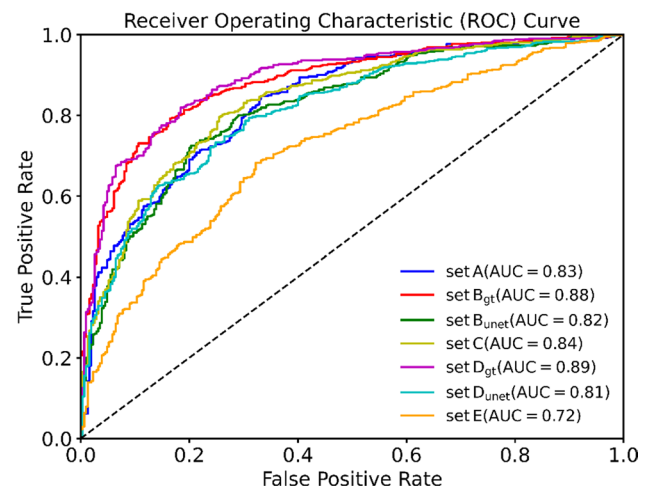


Fig. 3 Final receiver operating characteristic curves for all seven pseudo color image sets

Results

Figure 2 shows examples of the segmented masses. For each mass, both the ground truth (red) and automated segmentation results (green) are demonstrated. After fivefold cross validation, automated mass segmentation via a U-Net achieved a DSC of 0.894 ± 0.002 and a Jaccard Index of 0.814 ± 0.003 . Figure 3 demonstrates the ROC (receiver operation characteristic) curve of all proposed models, while Table 3 presents the six different performance metrics for each pseudo color image set. Overall, both pseudo color sets B_{gt} and D_{gt} outperform all

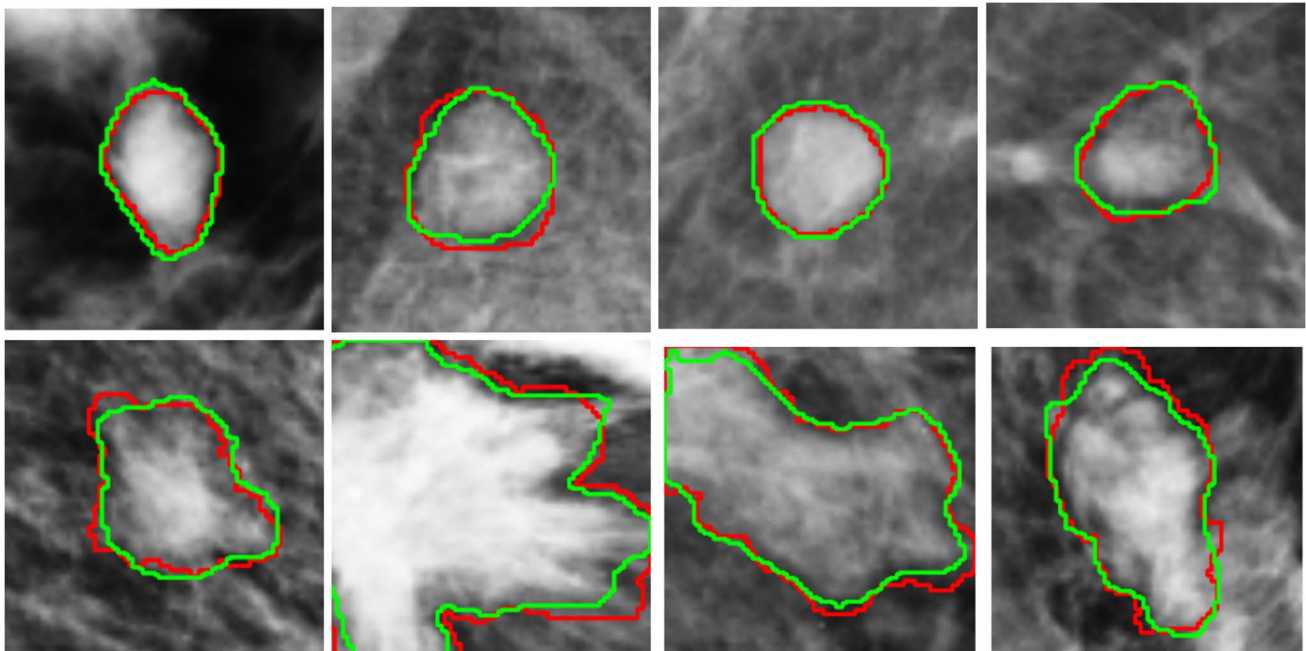
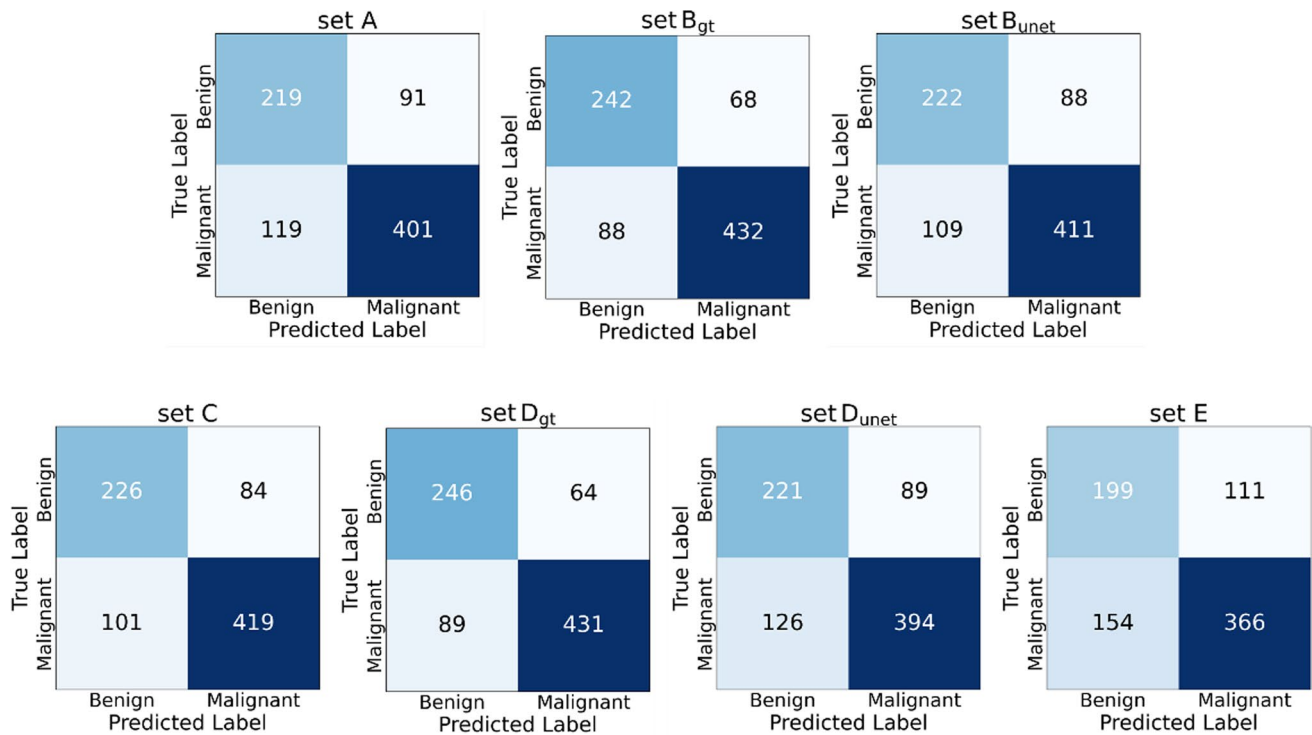


Fig. 2 Examples of the manual ground truth segmentation in red and the U-Net-produced segmentation mask in green. The top row is benign cases, and the bottom row is malignant cases

Table 3 Mean and standard deviation of all metrics computed from each test fold over fivefold cross-validation

	AUC	Acc	Precision	Sensitivity	Specificity	F1 score
<i>A</i>	0.833 ± 0.014	0.747 ± 0.030	0.817 ± 0.028	0.771 ± 0.068	0.706 ± 0.072	0.791 ± 0.033
<i>B_{gt}</i>	0.882 ± 0.012	0.812 ± 0.009	0.864 ± 0.016	0.831 ± 0.016	0.781 ± 0.032	0.847 ± 0.008
<i>B_{unet}</i>	0.820 ± 0.015	0.763 ± 0.022	0.825 ± 0.025	0.790 ± 0.035	0.716 ± 0.053	0.806 ± 0.019
<i>C</i>	0.836 ± 0.014	0.777 ± 0.028	0.833 ± 0.020	0.806 ± 0.037	0.729 ± 0.034	0.819 ± 0.025
<i>D_{gt}</i>	0.889 ± 0.012	0.816 ± 0.020	0.871 ± 0.021	0.829 ± 0.025	0.794 ± 0.039	0.849 ± 0.017
<i>D_{unet}</i>	0.812 ± 0.015	0.741 ± 0.0250	0.816 ± 0.0315	0.758 ± 0.017	0.713 ± 0.059	0.786 ± 0.018
<i>E</i>	0.718 ± 0.018	0.681 ± 0.035	0.767 ± 0.030	0.704 ± 0.039	0.642 ± 0.050	0.734 ± 0.031

**Fig. 4** Confusion matrices for each of the seven pseudo color sets. Within each matrix, the top left corner represents the number of true negative (TN) cases, the top right corner represents the number of

false positive (FP) cases, the bottom left corner represents the number of false negative (FN) cases, and the bottom right corner represents the number of true positive (TP) cases.

other sets in terms of AUC, accuracy, sensitivity, specificity, precision, and F1 score. These two sets are created using the ground truth segmentation masks. The estimated AUC values of sets *B_{gt}* and *D_{gt}* are significantly higher than sets *A*, *B_{unet}*, *C*, *D_{unet}*, and *E* ($p < 0.05$ for all comparisons). Meanwhile, there are no statistically significant differences between the AUC of set *B_{gt}* and set *D_{gt}* ($p = 0.6036$), and the performance of the pseudo color sets *B_{unet}* and *D_{unet}* is no better than sets *A*, *C*, or *E*. All sets perform significantly better than set *E* in terms of AUC ($p < 0.001$ for all comparisons).

Figure 4 shows the confusion matrix of all these different classification models. For the model trained with *B_{gt}*, a total of 330 cases were predicted as “negative,” among which 242 cases were confirmed as benign by the following biopsy examination, generating a positive predictive value of 0.733. Meanwhile, a total of 500 cases were predicted as “positive,” while 432 cases

are confirmed as true positive (i.e., malignant); thus, the negative predictive value (NPV) is achieved as 0.864. Combining all these results together, the overall prediction accuracy is generated as 0.812. Using the same method, the accuracy of the other six models was also evaluated, which are summarized in Table 3.

Discussion

In this study, we investigated a novel pseudo color image generation approach for deep transfer learning and evaluated its effectiveness in classifying suspicious breast lesions. We hypothesize that creating pseudo color images with additional morphological features will provide increased complementary information to a deep network pre-trained on the

ImageNet database. Overall, the results of this work support our hypothesis, which has the following unique contributions.

First, this research further advances the previously reported work in this field. In our previous study, we investigated the effects of creating pseudo color images using various pre-processing techniques to improve the performance of a CADx framework in classifying malignant and benign lesions [13]. In that study, the pseudo color image set was generated with variants of the original single-channel image that either suppressed noise or enhanced contrast. Similar studies were also conducted by Li et al. [35] and Min et al. [12]. In this work, we demonstrate the addition of the segmented mass to the pseudo color images as the input of the deep transfer learning model, which shows significant improvement in classifying malignant and benign lesions. To verify that only the segmented mass image in the pseudo color set is responsible for the increase in performance, we replaced one original mammogram in set D_{gt} with a histogram equalized image (set B_{gt}), and there is no significant difference ($AUC=0.889$ and 0.882 , $p=0.604$) between these two sets. In addition, when only using the preprocessed images (i.e., bilateral filter and histogram equalization, Set C), the classification performance did not improve as compared to the simply stacked image set. This can be attributed by the fact that the deep transfer model cannot effectively learn the correct mass morphological information, but can identify the information provided by the other preprocessing filters.

Second, this study indicates the importance of accurate mass segmentation. During the experiment, the manual segmentation mask is used as the ground truth image for the automated segmentation task. We recognize that acquiring a manual segmentation for every suspicious mass is a highly time-consuming and error-prone task. To overcome future challenges in obtaining ground truth segmentation, we employed a U-Net to demonstrate that a fully automated mass segmentation network can serve as a substitute for manual segmentation masks. However, we observed that the model performance did not improve or even slightly deteriorated when incorporating the mass contour information generated from the U-Net. This is because the automated segmentation result is not as accurate as the ground truth. In some cases, these differences are quite significant. The deep transfer model may even be distracted by this inaccurate information, which adversely impacts the final classification performance. Therefore, developing a high-accuracy mass segmentation scheme will be an important direction for pseudo color image generation.

While this proof-of-concept study provides valuable insights, it also has some notable limitations. First, the dataset used in this work has a limited number of cases which is acquired from a single institution. Thus, the generalizability and robustness of the proposed method should be evaluated and validated more comprehensively on a larger multi-site dataset containing more breast cancer patients. Second, in this study, only one kind of deep CNN model (i.e., VGG) is

utilized as the feature extractor. More different kinds of deep learning models [36], especially transformer-based models [37], should be adopted to test this pseudoscalar generation method. In addition, parameter-efficient training methods [38] should also be adopted to optimize the model. Third, as mentioned above, an accurate segmentation scheme should be developed for the application of the mass contour-based image set. Some latest transformer-based segmentation models [39], especially the CLIP (Contrastive Language-Image Pre-training)-based universal segmentation models [40, 41], should be utilized in the future study to enhance the segmentation accuracy. Fourth, this initial study only uses the suspicious masses which appear in both the CC and MLO view of the same breast, and all cases which fail in the ipsilateral matching step are also discarded. These discarded masses may be more difficult cases, but they were not tested in this proposed study. Despite these limitations, this study is valid and highly meaningful, as it is the first step to verify the significance and importance of the mass morphological information for deep learning-based classification model.

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Declarations

Conflict of Interest The authors declares no conflict of interest in this investigation.

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